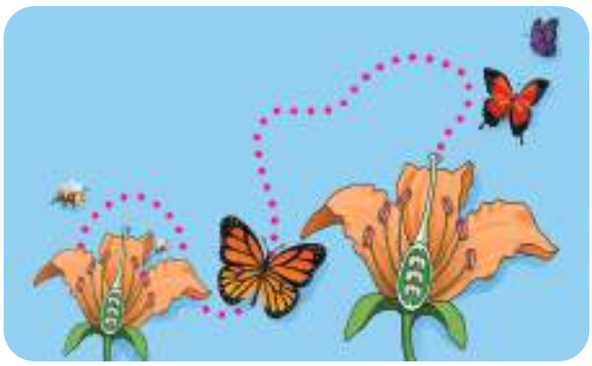
In the above experiment we transferred the pollen grains from male flower to the female flower. This is called as an **artificial pollination**. However, in nature there are many ways in which pollen grains reach the stigma of the flower and it is called as **natural pollination.**

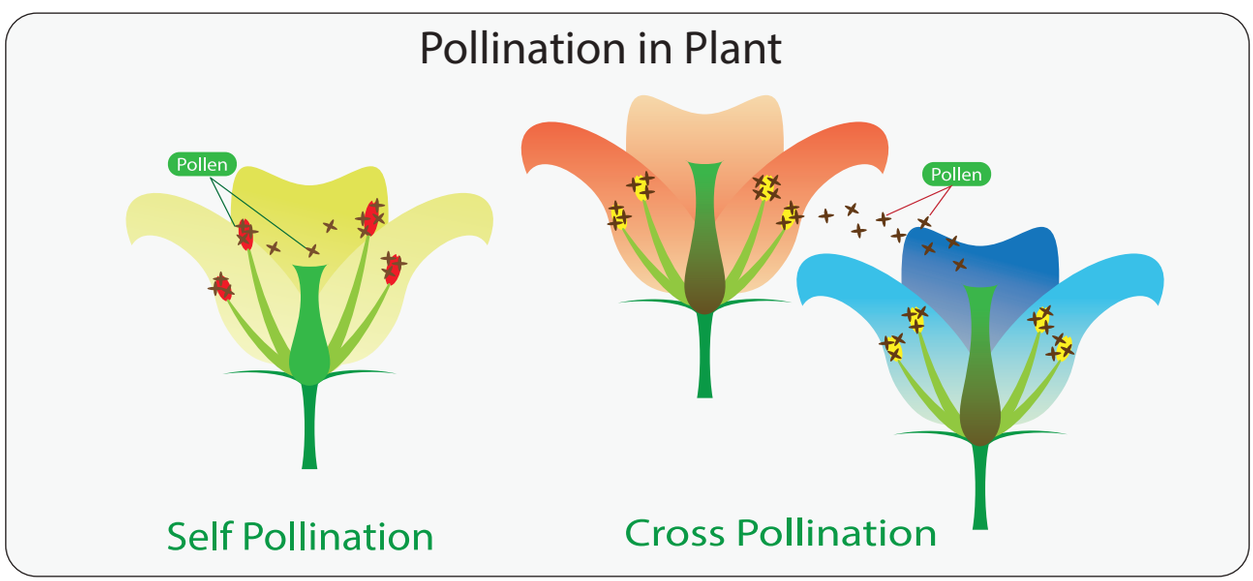


In some plants like grasses, pollen grains are light. Stamens shed pollen grains, and are carried by wind to other flower. Insects, birds are also agents of pollination. Bees, butterflies and variety of birds hover around flowers. They help to carry pollen from one flower to another. Pollen grains stick to their legs, wings or abdomen when they move from one flower to another. This is called as **cross pollination**

When you shake stamens, pollen grains fall. Thus, when wind shakes the flower or when a butterfly agitates the flower, pollen grains could fall onto the stigma of the same flower. Some plants that have both the male and female parts within a single flower (bisexual) pollinate by this means. This is called as **self pollination**.

**Table 5.1 Differences between self pollination and cross pollination.**

| Self Pollination | Cross Pollination |
|---|---|
| Pollen grains are transferred from the anther to the stigma of the same flower. | Pollen grains are transferred from the anther of one flower to the stigma of another flower of the same plant or same variety. |
| Plants do not need to produce pollen grains in a large quantity for self pollination | Plants need to produce pollen grains in larger quantities to increase the chance of pollination. |
| It does not produce changes in the characteristics of new plants. | Cross pollination does introduce variations in the characteristics of new plants. |



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